

Family 13: From The Original Table To The Modified Table

1. Setup

Structure used for the check

- Carbon nanotube $(5,0)$
- family = 13
- nrot = 10
- a = 4.26 Å
- α_U is the azimuthal angle of generator U , β_S is the azimuthal angle of the mirror plane generator σ_V :

$$\alpha_U \approx -\frac{\pi}{10}, \quad \beta_S \approx -\frac{2\pi}{10}.$$

Three q-points used

- Gamma:

$$q = 0$$

- a generic nonzero q-point:

$$q = 0.25 \cdot \frac{2\pi}{a}$$

- Brillouin-zone boundary:

$$q = 0.5 \cdot \frac{2\pi}{a} = \frac{\pi}{a}$$

2. The Two Tests

2.1 Orthogonality of irreps

In finite group, the characters must satisfy

$$\frac{1}{|G|} \sum_{g \in G} \chi_\mu(g) \chi_\nu(g)^* = \delta_{\mu\nu}.$$

More strongly, the irrep matrix elements must satisfy Schur orthogonality:

$$\frac{d_\mu}{|G|} \sum_{g \in G} D_{ij}^{(\mu)}(g) D_{kl}^{(\nu)}(g)^* = \delta_{\mu\nu} \delta_{ik} \delta_{jl}.$$

2.2 Idempotency of the Bloch projector

The projector used by the code is:

$$P_\mu = \frac{d_\mu}{|H_k|} \sum_{h \in H_k} \chi_\mu(h)^* D_{\text{Bloch}}(h)$$

The required condition is

$$P_\mu^2 = P_\mu.$$

Numerically, the relevant error measure is

$$\frac{\|P_\mu^2 - P_\mu\|}{\|P_\mu\|}.$$

3. First Failure: The Original Gamma Table Is Not A Set Of Orthogonal Irreps

Using the original table at Gamma:

- number of irreps: 26
- dimension pattern:

$$1, 1, 1, 1, \underbrace{2, \dots, 2}_{18 \text{ times}}, 1, 1, 1, 1$$

This already signals a problem. For a finite group, the irreps must satisfy

$$\sum_{\mu} d_{\mu}^2 = |G|.$$

Here the full line-group factor used at Gamma has $|G| = 40$, but the original table gives

$$8 \times 1^2 + 18 \times 2^2 = 80,$$

which is impossible. So the original table double-counts irreps.

Numerical orthogonality result at Gamma

Original table:

- character orthogonality max off-diagonal error: 0.7233833157466387
- irrep-matrix orthogonality max off-diagonal error: 0.6874954178467586
- projector max idempotency error: 0.8013650440433105

Modified table:

- character orthogonality max off-diagonal error: 4.327696520090019e-07
- irrep-matrix orthogonality max off-diagonal error: 3.217625350053425e-07
- projector max idempotency error: 8.978752676919702e-06

So at Gamma, the modified table is self-consistent and the original one is not.

What this forces

This already forces two structural changes:

1. the m range cannot run over both m and $n-m$,
2. the $m = n/2$ branch cannot stay in the same 2D sector as generic $0 < m < n/2$.

4. The m Range Must Be Reduced To $0 \leq m \leq \text{floor}(n/2)$

The original Family 13 implementation effectively double-counted sectors related by

$$m \leftrightarrow n - m.$$

For generic m , the $(+m, -m)$ pairing is already built into the 2D or 4D representation matrices. So if both m and $n-m$ are enumerated as independent rows, the same physical irrep is counted twice.

That is why the current implementation uses

$$m = 0, 1, \dots, \left\lfloor \frac{n}{2} \right\rfloor$$

for all q values.

5. Why The Rotation Phases Must Be Doubled

The original table used phases of the form

$$e^{im\pi/n}$$

for the screw generator, and

$$e^{i2m\pi/n}$$

for the pure rotation column.

But the Family 13 screw generator rotates by

$$\frac{2\pi}{n},$$

not by π/n . Therefore an angular-momentum label m must pick up

$$e^{im(2\pi/n)} = e^{i2m\pi/n}.$$

Likewise, the pure rotation generator used in the table corresponds to twice that angle, so it must carry

$$e^{i4m\pi/n}.$$

Without this correction, different m sectors are assigned the wrong phases and fail to separate orthogonally.

One explicit Gamma-point symptom in the original table was the large overlap between the two rows with:

- $m = 1$, $\text{piH} = -1$
- $m = 2$, $\text{piH} = -1$

Their character overlap was

$$0.7233833157466387,$$

which is far from zero for inequivalent irreps.

So the phase doubling is forced by orthogonality.

6. Why $m = n/2$ Must Split Off At Gamma

For even n , the label

$$m = \frac{n}{2}$$

is special because

$$m \equiv -m \pmod{n}.$$

So the usual 2D pairing of $+m$ and $-m$ collapses: those are no longer two distinct sectors.

If one still keeps the old 2D branch at $m=n/2$, the representation becomes reducible.

That is exactly what the Gamma-point counting shows.

With $n=10$, if $m=5$ were kept in the 2D branch, the table would still overcount:

$$4 \times 1^2 + 10 \times 2^2 = 44 > 40.$$

So at Gamma:

- $m = 0$ stays in the 1D func0 branch,
- $m = n/2$ must also move into the 1D func0 branch,
- only $0 < m < n/2$ remain in the 2D func1 branch.

This is precisely the current branch logic in the modified file.

So the Gamma-point row that was previously treated like a generic 2D E_m row must be replaced by a 1D row of the B_0 type.

7. Why The Generic-q $m = 0$ or $n/2$ Sector Must Use ${}_k E_m^{\Pi_V}$

At nonzero q , the correct projector is built from the k -preserving little group. In this convention, the natural 2D pairing is:

$$(+k, -k),$$

not $(+m, -m)$.

That means:

- for $m = 0$, the relevant nontrivial doubling is the k sector,
- for $m = n/2$, exactly the same is true, because $+m$ and $-m$ are no longer independent.

Therefore both $m=0$ and $m=n/2$ must live in the same func2 -type 2D branch at nonzero q .

If the old table keeps $m=n/2$ in the 4D branch, the representation is artificially doubled and fails the little-group check.

Numerical symptom in the original table

At generic nonzero q , on the actual little-group subset used by the code:

Original table:

- maximum off-block mixing error: 2.0
- little-group character orthogonality max off-diagonal error: 0.6077683537175255
- little-group matrix orthogonality max off-diagonal error: 0.6392453221499665

Modified table:

- maximum off-block mixing error: 0.0
- little-group character orthogonality max off-diagonal error: $3.9730721768184846e-16$
- little-group matrix orthogonality max off-diagonal error: $6.4796043696850615e-16$

So the little-group test forces the $m=0/n/2$ sector into the reduced 2D func2 branch.

8. Why The Old Off-Diagonal Block In ${}_k E_m^{A/B}$ (the new ${}_k E_m^{\Pi_V}$)

The old func2 branch used an off-diagonal matrix of the form

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

for one of the non-rotational generators.

But in the nonzero- q projector logic, the relevant basis is the $(+k, -k)$ basis, and the code explicitly identifies little-group elements as those whose irrep matrices are block-diagonal in that basis.

If the generator wrongly swaps $+k$ and $-k$, then:

- it no longer belongs to the k -preserving block used by the projector,
- the off-block norm becomes large,
- the induced projector ceases to be idempotent.

This is exactly what happens in the original table for the generic- q $m=0$ rows:

- largest off-block error: 2.0
- projector idempotency error: 0.7515195920480531

Replacing that off-diagonal matrix by the identity restores:

- zero off-block leakage on the actual little group,
- projector idempotency at the 1e-5 level.

So this correction is forced by the nonzero-q little-group structure.

9. Why The BZ-Boundary $m = n/2$ Row Must Use $k E_m^{A/B}$ Instead Of $\Pi E_{n/2}^{\Pi_U}$ (merging to $k E_m^{\Pi_V}$ in the new table)

This is the cleanest and strongest step in the derivation.

At the Brillouin-zone boundary,

$$q = \frac{\pi}{a}.$$

For the screw operation, the translation part is $a/2$, so the Bloch phase in $D_{\text{Bloch}}(g)$ is

$$e^{iq(a/2)} = e^{i(\pi/a)(a/2)} = e^{i\pi/2} = i.$$

But the old $\Pi E_{n/2}^{\Pi_U}$ branch is purely real:

$$\pm 1$$

only. It contains no q-dependent phase at all.

Therefore $\Pi E_{n/2}^{\Pi_U}$ cannot be the correct character source for the projector in the current code convention, because the projector uses Bloch matrices that already include the factor $e^{i q \tau_z}$.

Numerical evidence

At the BZ boundary, original table:

- the two $m=n/2$ rows using the old $\Pi E_{n/2}^{\Pi_U}$ branch give projector idempotency error

$$\approx 0.8426.$$

Modified table:

- the corresponding two $m=n/2$ rows, now using $k E_m^{A/B}$, give projector idempotency errors

$$6.7 \times 10^{-6} \quad \text{and} \quad 9.5 \times 10^{-6}.$$

This is not a small improvement. It is the difference between “not a projector” and “numerically a projector”.

So in the current Bloch-phase-inclusive convention, the BZ-boundary $m=n/2$ row must use $k E_m^{A/B}$, not $\Pi E_{n/2}^{\Pi_U}$.

10. Why alpha_U And beta_S Appear

The phases `alpha_u` and `beta_s` are not arbitrary fit parameters. They are extracted from the actual geometry of the detected generators by `_extract_generator_angles()` in:

- `src/pulgon_tools/symmetry_projector.py`

and then inserted into the irrep formulas.

This is needed because the actual `c_2'` axis and mirror plane are not generally aligned with a fixed x-axis reference.

In the basis $(|m\rangle, |-m\rangle)$, a generator with geometric angle `alpha_u` or `beta_s` naturally contributes phases like

$$e^{\pm i 2 m \alpha_U}, \quad e^{\pm i 2 m \beta_S}.$$

Numerical evidence

If the modified table is evaluated with the actual extracted angles, the Gamma-point character orthogonality error is:

$$4.3 \times 10^{-7}.$$

If the same modified formulas are evaluated with those angles forcibly set to zero, the Gamma-point maximum off-diagonal character overlap becomes:

$$0.25.$$

So the geometry-dependent phases are needed for full consistency with the actual detected generators.

11. Key Numerical Summary

Gamma

- original: character off-diagonal 0.723 , projector error 0.801
- modified: character off-diagonal 4.3e-7 , projector error 9.0e-6

Generic nonzero q, actual little group

- original: little-group character off-diagonal 0.608 , projector error 0.839
- modified: little-group character off-diagonal 4e-16 , projector error 9.5e-6

BZ boundary

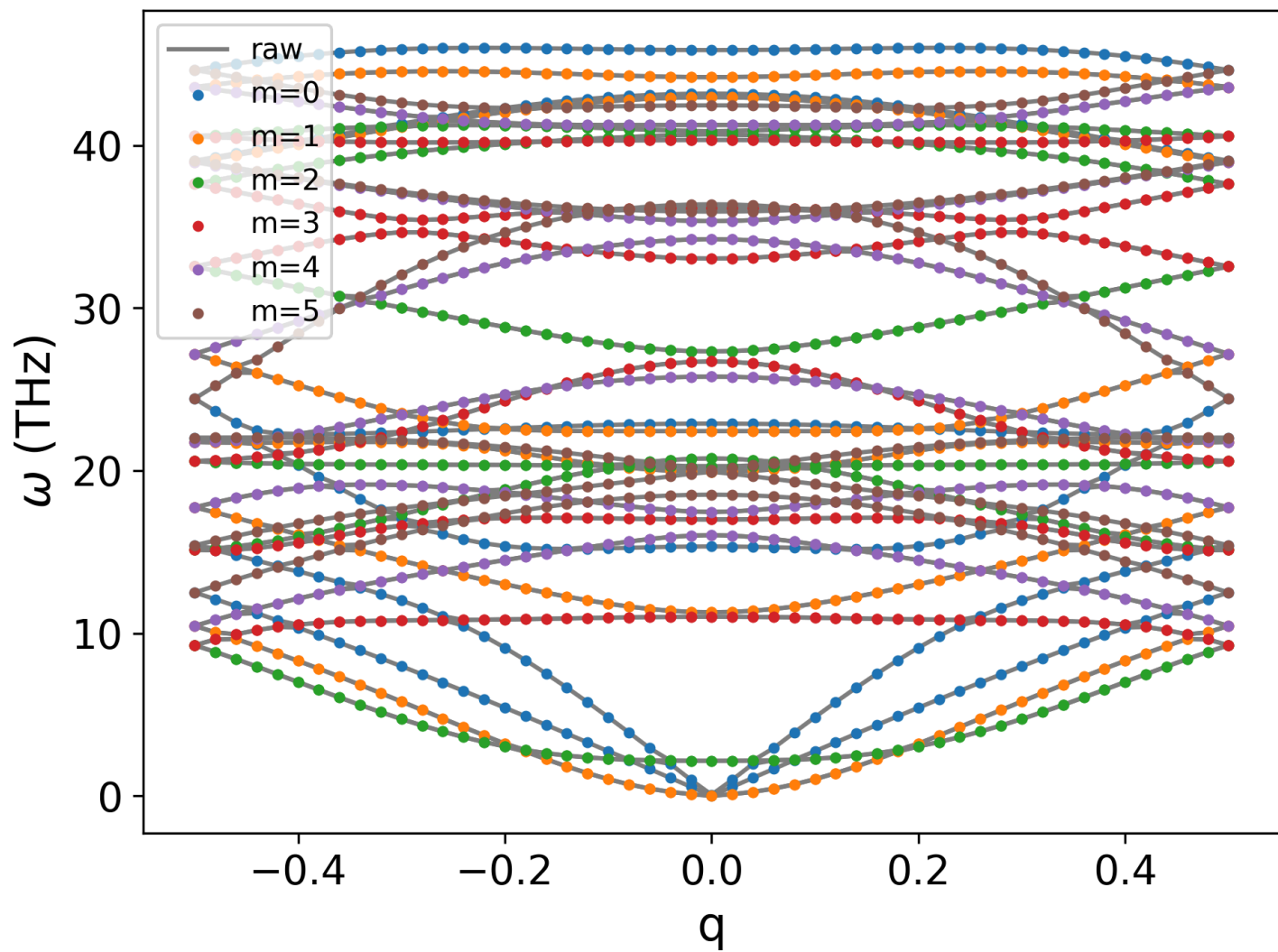
- original: worst projector error 0.847
- modified: worst projector error 9.5e-6

These are the numbers that rule out the old table and validate the new one.

12. Testing sym-adapted-matrix by Phonon

Carbon nanotube (5,0)

Irreps by m



Full irreps

